

Alps Alpine ThumbPointer™ HMI Board (Qwiic) - Application Notes

These are the application notes for the Alps Alpine ThumbPointer™ HMI Board (Qwiic).

I2C Address Assignments

A0	A1	I2C Address
0	0	0x62 (Default)
0	1	0x63
1	0	0x64
1	1	0x65

Message Sequence & Format

All Alps Alpine QWIIC Sensor/HMI boards use a similar I²C communication sequence between the master and slave. In these exchanges, the Alps Alpine QWIIC Sensor/HMI boards always functions as the slave device.

The master is responsible for initiating both read and write operations. To change a setting or request data from a board, follow the sequence below:

**** Placeholder Sequence. Add diagram later ****

- The Master must first send request message → The board will prepare the data or change setting as requested
- The Master must then send read message with size required for the response → The board will send back the requested data response

Request Message Types

The following message types are used to indicate whether the board should prepare requested data for the user to read back or execute an operation. This message type ID is located at the beginning of the message payload sent from the I2C master device to the Alps Alpine Sensor/HMI board.

Request type	Message ID Value (uint8)	Description
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Request Data	0x01	Used to request the board to prepare data that is in the data request (proceeding value in the request payload)
Request Operation	0x02	Used to request the board to execute a operation (proceeding operation in the request payload)

Request Data Types & Formats (0x01):

The following are different request data types that are supported by the Alps Alpine ThumbPointer™ HMI Board (Qwiic) and their expected formats.

“WHO AM I” Data Request Message (0x01):

The “WHO AM I” message can be used to both determine if a Alps Alpine Sensor/HMI board is present and can be used to determine what type of Sensor/HMI board it is.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x01	Who Am I Data Request

Response I2C Payload:

During response request for “WHO AM I”, Master must perform a read request with **3 bytes**.

Byte Index	Value	Description
0	0x01	Message Header
1	0x01	Message Header
2	0x01	Device type indicator. 0x01 = Alps Alpine ThumbPointer™ HMI Board (Qwiic)

“Get Formatted Values” Data Request Message (0x02):

The “Get formatted values” message, can be used to request the current values from the HMI board in a more friendly format for reading. Response data and its format may change based on current set modes. In the formatted request, the HMI board presents to the X & Y axis as location from the center point. The board also flips the value for the button press to be more of a “is pressed” representation.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x02	Request Formatted Values

Response I2C Payload:

In default operating mode, during the response request for “Get Formatted Values Data”, the Master must perform a read request with **9 bytes**.

Byte Index	Content	Description
0 – 3	<code>formattedXAxis</code> (<code>float</code> , little-endian)	*The x-axis formatted reading (-1 to 1). Center = 0.0
4 – 7	<code>formattedYAxis</code> (<code>float</code> , little-endian)	*The y-axis formatted reading (-1 to 1). Center = 0.0
8	<code>isSwitchPressed</code> (<code>uint8_t</code>)	<code>0x01</code> if pressed, <code>0x00</code> if not pressed

“Get Raw Values” Data Request Message (0x03):

The “Get Raw values” message, can be used to request the current raw data that the HMI board reads from the Alps Alpine component. Response data and its format may change based on current set modes.

Request I2C Payload:

Byte Index	Value	Description
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0	0x01	Request Data type
1	0x03	Request Raw Values

Response I2C Payload:

In default operating mode, during the response request for “Get Raw Values”, the Master must perform a read request with **5 bytes**. Each value set represents raw analog and digital readings from the ThumbPointer™s pins.

Byte Index	Content	Description
0 – 1	<code>xAxisRead</code> (<code>uint16_t</code> , little-endian)	*The x-axis reading (0–4095)
2 – 3	<code>yAxisRead</code> (<code>uint16_t</code> , little-endian)	*The y-axis reading (0–4095)
4	<code>switchPressedRaw</code> (<code>uint8_t</code>)	<code>0x00</code> if pressed, <code>0x01</code> if not pressed

“Get Is Data Ready” Data Request Message (0x04):

The “Get Is Data Ready” message can be used to check whether the HMI board is ready to provide component data readings when requested. It is good practice to perform this request after powering on the board and before requesting any data, continuing until the board indicates that it is ready. This message should also be used after any operation request to wait until the board is ready before sending additional requests.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x04	Request Is Data Ready Value

Response I2C Payload:

In default operating mode, during the response request for “Get Is Data Ready”, the Master must perform a read request with **1 byte**.

Byte Index	Content	Description
0	isDataReady (uint8_t)	• 0x00 = Data Not Ready • 0x01 = Data Ready

“Get Firmware Version” Data Request Message (0x05):

The “Get Firmware Version” message can be used to check the firmware version of the HMI board.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x05	Request Firmware Version

Response I2C Payload:

In default operating mode, during the response request for “Get Firmware Version”, the Master must perform a read request with **2 bytes**.

Byte Index	Content	Description
0	MAJOR_VERSION (uint8_t)	The firmwares major value
1	MINOR_VERSION (uint8_t)	The firmwares minor value

Request Operation Types & Formats (0x02):

The following are different request operation types that are supported by the Alps Alpine ThumbPointer™ HMI Board (Qwiic) and their expected formats. These requests do not have any responses in return.

These requests do not have any responses in return.

“Perform Build-In Calibration” Operation Request Message (0x01):

The “Perform Build-In Calibration” message can be used to request the HMI board to perform the built in calibration operation. This can be used to zero out the center position of the ThumbPointer™ component. ***This calibrated value is applied as a offset value to both formatted and raw reading values.** On power up, the boards calibration values are reset to default 0 for both direction (meaning no calibration is applied).

****This should only be performed with the ThumbPointer™ in the center position ****

Request I2C Payload:

Byte Index	Value	Description
0	0x02	Request Operation type
1	0x01	Request Perform Built-In Calibration

“Enter DFU Mode” Operation Request Message (0xff):

The “Enter DFU Mode” message can be used to request the HMI board to enter its DFU mode. This allows for a external I2C Master device to perform DFU over I2C interface.

Request I2C Payload:

Byte Index	Value	Description
0	0x02	Request Operation type
1	0xff	Request Enter DFU Mode